

数学物理方法 2

第一次作业

1.

$$\begin{aligned}\vec{r} = r\vec{e}_r &\implies \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\vec{e}_r) = \frac{dr}{dt}\vec{e}_r + r\frac{d\vec{e}_r}{dt} = \frac{dr}{dt}\vec{e}_r + r\frac{d\theta}{dt}\vec{e}_\theta \\ \frac{d \text{ vecv}}{dt} &= \frac{d}{dt}\left(\frac{d\vec{r}}{dt}\right) = \frac{d}{dt}\left(\frac{dr}{dt}\vec{e}_r + r\frac{d\theta}{dt}\vec{e}_\theta\right) = \frac{d^2r}{dt^2}\vec{e}_r + \frac{dr}{dt}\frac{d\theta}{dt}\vec{e}_\theta + \frac{dr}{dt}\frac{d\theta}{dt}\vec{e}_\theta + r\frac{d^2\theta}{dt^2}\vec{e}_\theta + r\frac{d\theta}{dt}\frac{d\vec{e}_\theta}{dt} \\ &= \frac{d^2r}{dt^2}\vec{e}_r + 2\frac{dr}{dt}\frac{d\theta}{dt}\vec{e}_\theta + r\frac{d^2\theta}{dt^2}\vec{e}_\theta - r\left(\frac{d\theta}{dt}\right)^2\vec{e}_r\end{aligned}$$

2.

$$(1) \nabla \frac{1}{r} = \frac{\partial}{\partial x}\vec{e}_x + \frac{\partial}{\partial y}\vec{e}_y + \frac{\partial}{\partial z}\vec{e}_z = \frac{r_x\vec{e}_x}{(r_x^2+r_y^2+r_z^2)^{3/2}} + \frac{r_y\vec{e}_y}{(r_x^2+r_y^2+r_z^2)^{3/2}} + \frac{r_z\vec{e}_z}{(r_x^2+r_y^2+r_z^2)^{3/2}} = \frac{\vec{r}}{r^3}$$

$$(2) \nabla \times \frac{\vec{r}}{r^3} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{x}{r^3} & \frac{y}{r^3} & \frac{z}{r^3} \end{vmatrix} = (-3\frac{yz}{r^5} + 3\frac{yz}{r^5})\vec{e}_x - (-3\frac{xz}{r^5} + 3\frac{xz}{r^5})\vec{e}_y + (-3\frac{xy}{r^5} + 3\frac{xy}{r^5})\vec{e}_z = \vec{0}$$

$$(3) \nabla \times \vec{r} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \vec{0}$$

3.

我们假设有如下坐标变换: $\begin{cases} r = \sqrt{x^2 + y^2 + z^2} \\ \tan\theta = \frac{\sqrt{x^2+y^2}}{z} \\ \tan\varphi = \frac{x}{y} \end{cases}$, 则可以导出如下的偏微分关系:

$$\begin{cases} \frac{\partial}{\partial x} = \frac{\partial}{\partial r}\frac{\partial r}{\partial x} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial x} + \frac{\partial}{\partial \varphi}\frac{\partial \varphi}{\partial x} = \sin\theta\sin\varphi\frac{\partial}{\partial r} + \frac{\cos\theta\sin\varphi}{r}\frac{\partial}{\partial \theta} + \frac{\cos\varphi}{r\sin\theta}\frac{\partial}{\partial \varphi} \\ \frac{\partial}{\partial y} = \frac{\partial}{\partial r}\frac{\partial r}{\partial y} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial y} + \frac{\partial}{\partial \varphi}\frac{\partial \varphi}{\partial y} = \sin\theta\cos\varphi\frac{\partial}{\partial r} + \frac{\cos\theta\cos\varphi}{r}\frac{\partial}{\partial \theta} - \frac{\sin\varphi}{r\sin\theta}\frac{\partial}{\partial \varphi} \\ \frac{\partial}{\partial z} = \frac{\partial}{\partial r}\frac{\partial r}{\partial z} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial z} + \frac{\partial}{\partial \varphi}\frac{\partial \varphi}{\partial z} = \cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial \theta} \end{cases}$$

$$\begin{aligned}\text{因此拉普拉斯算子为 } \Delta &= \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} = (\sin\theta\sin\varphi\frac{\partial}{\partial r} + \frac{\cos\theta\sin\varphi}{r}\frac{\partial}{\partial \theta} + \frac{\cos\varphi}{r\sin\theta}\frac{\partial}{\partial \varphi})^2 + (\sin\theta\cos\varphi\frac{\partial}{\partial r} + \frac{\cos\theta\cos\varphi}{r}\frac{\partial}{\partial \theta} - \frac{\sin\varphi}{r\sin\theta}\frac{\partial}{\partial \varphi})^2 + (\cos\theta\frac{\partial}{\partial r} - \frac{\sin\theta}{r}\frac{\partial}{\partial \theta})^2 \\ &= (\sin^2\theta\sin^2\varphi + \sin^2\theta\cos^2\varphi + \cos^2\theta)\frac{\partial^2}{\partial r^2} + (\sin^2\varphi\cos\theta\sin\theta + \cos^2\varphi\cos\theta\sin\theta - \cos\theta\sin\theta)\frac{2}{r}\frac{\partial^2}{\partial r\partial \theta} + \\ &\quad (\sin\varphi\cos\varphi - \sin\varphi\cos\varphi)\frac{2}{r}\frac{\partial^2}{\partial r\partial \varphi} + (\cos^2\theta\sin^2\varphi + \cos^2\theta\cos^2\varphi + \sin^2\theta)\frac{1}{r^2}\frac{\partial^2}{\partial \theta^2} + \\ &\quad (\cot\theta\cos\varphi\sin\varphi - \cot\theta\cos\varphi\sin\varphi)\frac{2}{r^2}\frac{\partial^2}{\partial \theta\partial \varphi} + \frac{\cos^2\varphi + \sin^2\varphi}{r^2\sin^2\theta}\frac{\partial^2}{\partial \varphi^2} - \frac{\sin^2\varphi\cos\theta\sin\theta}{r^2}\frac{\partial}{\partial \theta} - \frac{\cos\varphi\sin\varphi}{r^2}\frac{\partial}{\partial \varphi} + \frac{\cos^2\theta\sin^2\varphi}{r}\frac{\partial}{\partial r} - \\ &\quad \frac{\cos\theta\sin\theta\cos^2\varphi}{r^2}\frac{\partial}{\partial \theta} - \frac{\cos^2\theta\sin\varphi\cos\varphi}{r^2\sin^2\theta}\frac{\partial}{\partial \varphi} + \frac{\cos^2\varphi}{r}\frac{\partial}{\partial r} + \frac{\cos\theta\cos^2\varphi}{r^2\sin\theta}\frac{\partial}{\partial \theta} - \frac{\sin\varphi\cos\varphi}{r^2\sin^2\theta}\frac{\partial}{\partial \varphi} - \frac{\cos^2\varphi\cos\theta\sin\theta}{r^2}\frac{\partial}{\partial \theta} + \frac{\sin\varphi\cos\varphi}{r^2}\frac{\partial}{\partial \varphi} +\end{aligned}$$

$$\frac{\cos^2\theta\cos^2\varphi}{r}\frac{\partial}{\partial r}-\frac{\cos^2\varphi\cos\theta\sin\theta}{r^2}\frac{\partial}{\partial\theta}+\frac{\cos^2\theta\cos\varphi\sin\varphi}{r^2\sin^2\theta}\frac{\partial}{\partial\varphi}+\frac{\sin^2\varphi}{r}\frac{\partial}{\partial r}+\frac{\cos\theta\sin^2\varphi}{r^2\sin\theta}\frac{\partial}{\partial\theta}+\frac{\sin\varphi\cos\varphi}{r^2\sin^2\theta}\frac{\partial}{\partial\varphi}+\frac{\cos\theta\sin\theta}{r^2}\frac{\partial}{\partial\theta}+\frac{\sin^2\theta}{r}\frac{\partial}{\partial r}+\frac{\cos\theta\sin\theta}{r^2}\frac{\partial}{\partial\theta}$$

$$\text{化简得:}\Delta=\frac{\partial^2}{\partial r^2}+\frac{1}{r^2}\frac{\partial^2}{\partial\theta^2}+\frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}+\frac{2}{r}\frac{\partial}{\partial r}+\frac{\cos\theta}{r^2\sin\theta}\frac{\partial}{\partial\theta}.$$

4.

考虑分析在原长坐标中 x 至 $x+dx$ 一段杆内部的张力 F . 这一段杆的伸长量为 $u(x+dx, t) - u(x, t)$.

由 Hooke 定律, $\frac{F(x, t)}{S} = Y \frac{u(x+dx, t) - u(x, t)}{dx} = Y u_x(x, t) \implies F(x, t) = Y S u_x(x, t)$.

然后考虑 $x \sim x+dx$ 一段杆微元在两个端面分别受到 $F(x, t)$ (左侧微元对该微元的力, 即 $x-dx_- \sim x$ 内部的张力) 和 $F(x+dx, t)$ (右侧微元对该微元的力, 即 $x+dx \sim x+dx+dx_+$ 内部的张力). 其中 $F(x, t)$ 方向向左, $F(x+dx, t)$ 方向向右. 该段微元的质量为 $\rho A dx$. 因此由牛顿第二定律, 我们有

$\rho A dx * u_{tt}(x, t) = F(x+dx, t) - F(x, t) \implies \rho A u_{tt}(x, t) = F_x(x, t)$. 代入上面求出的张力表达式, 我们有:

$\rho A u_{tt}(x, t) = Y S u_{xx}(x, t) \implies \rho u_{tt}(x, t) = Y u_{xx}(x, t)$. 此即纵波波动方程. 边界条件为 $u(0, t) = 0$.

5.

傅里叶变换:

$$F(\omega) = \int_{-\infty}^{+\infty} \cos^2(x) e^{-i\omega x} dx = \int_{-\infty}^{+\infty} \frac{1+\cos 2x}{2} e^{-i\omega x} dx = \pi \delta(\omega) + \pi \delta(\omega+2) + \pi \delta(\omega-2)$$

傅里叶展开:

$$a_0 = \frac{1}{\pi} \int_0^\pi \cos^2(x) dx = \frac{1}{2}$$

$$a_n = \frac{2}{\pi} \int_0^\pi \cos^2(x) \cos 2nx dx = \frac{1}{\pi} \int_0^\pi (\cos 2nx + \frac{1}{2} \cos(2(n+1)x) + \frac{1}{2} \cos(2(n-1)x)) dx = \frac{1}{2} \delta_{1n}$$

$$b_n = \frac{2}{\pi} \int_0^\pi \cos^2(x) \cos 2nx dx = 0$$

$$\text{因此 } \cos^2(x) = \frac{1+\cos 2x}{2}$$

对比: 由于该函数在频域中是离散点谱, 因此实际上傅里叶变换和傅里叶展开给出的结果是相同的. 傅里叶变换得到的 $\delta(\omega)$ 对应展开式中的 $\frac{1}{2}$, $\delta(\omega-2)$, $\delta(\omega+2)$ 对应 $\cos 2x$.

6.

(1)

当 $f'(0) = 0$ 时, 考虑 $f(t)$ 的偶延拓 $f(t) := f(-t) (t < 0)$ 此时 $F(\omega) = \int_0^{+\infty} f(t) e^{-i\omega t} dt + \int_0^{+\infty} f(t) e^{i\omega t} dt = \int_0^T h e^{-i\omega t} dt + \int_0^T h e^{i\omega t} dt = \frac{2h}{\omega} \sin(\omega T)$

$$\text{因此 } f(t) = \frac{h}{\pi} \int_{-\infty}^{+\infty} \frac{\sin(\omega T)}{\omega} e^{i\omega t} d\omega$$

(2)

当 $f(0) = 0$ 时, 考虑 $f(t)$ 的奇延拓 $f(t) := -f(-t) (t < 0)$ 此时 $F(\omega) = \int_0^{+\infty} f(t) e^{-i\omega t} dt - \int_0^{+\infty} f(t) e^{i\omega t} dt = \int_0^T h e^{-i\omega t} dt - \int_0^T h e^{i\omega t} dt = \frac{2ih}{\omega} \cos(\omega T)$

因此 $f(t) = \frac{ih}{\pi} \int_{-\infty}^{+\infty} \frac{\cos(\omega T)}{\omega} e^{i\omega t} d\omega$

7.

由初始条件, 等式两端同时做拉普拉斯变换得 (记 $L[T] = F(p), L[g] = G(p)$):

$$p^2 F - pT(0) - T'(0) + \omega^2 a^2 F = p^2 F + \omega^2 a^2 F = G \implies F = G \frac{1}{p^2 + \omega^2 a^2}$$

由卷积定理, $T(t) = L^{-1}[G \frac{1}{p^2 + \omega^2 a^2}] = \int_0^t g(t') \frac{\sin(\omega a(t-t'))}{\omega a} dt' = \frac{1}{\omega a} g(t) * \sin(\omega a t)$

8.

记微分算子 $L = \frac{d^2}{dt^2} + (1-t)\frac{d}{dt} + \lambda$. 对于多项式 $P(t) = \sum a_n t^n, L[P] = \sum (\lambda a_n - n a_n + (n+1)a_{n+1} + (n+1)(n+2)a_{n+2})t^n$.

由于幂函数的线性无关性, 我们要求 $\forall n, (\lambda - n)a_n + (n+1)a_{n+1} + (n+1)(n+2)a_{n+2} = 0$ 有非零解 (这是一个线性方程组), 即要求该线性方程组的系数矩阵 $A(\lambda, n)$ 的行列式为 0. 我们不妨将 λ 提出来. 设 $A(\lambda, n) = \lambda I_n - \sum (i E_{i,i} - (i+1)E_{i,i+1} - (i+1)(i+2)E_{i,i+2}) = \lambda I_n - B_n$ ($E_{i,j}$ 表示 i, j 位置上为 1 的矩阵). 显然所有可能的 $\lambda, s.t. \det(\lambda I_n - B_n) = 0$ 对应着 B_n 的所有特征值.

由于 B_n 显然是一个上三角矩阵, 因此 B_n 的特征值即为对角线上的元素, 即 $1, 2, \dots, n$. 因此仅当 $\lambda \in N_+$ 时可能使得拉盖尔方程的解取为多项式. 特别地, 若想使解为 n 阶及以下多项式, 则需取 $\lambda = 1, 2, \dots, n$.